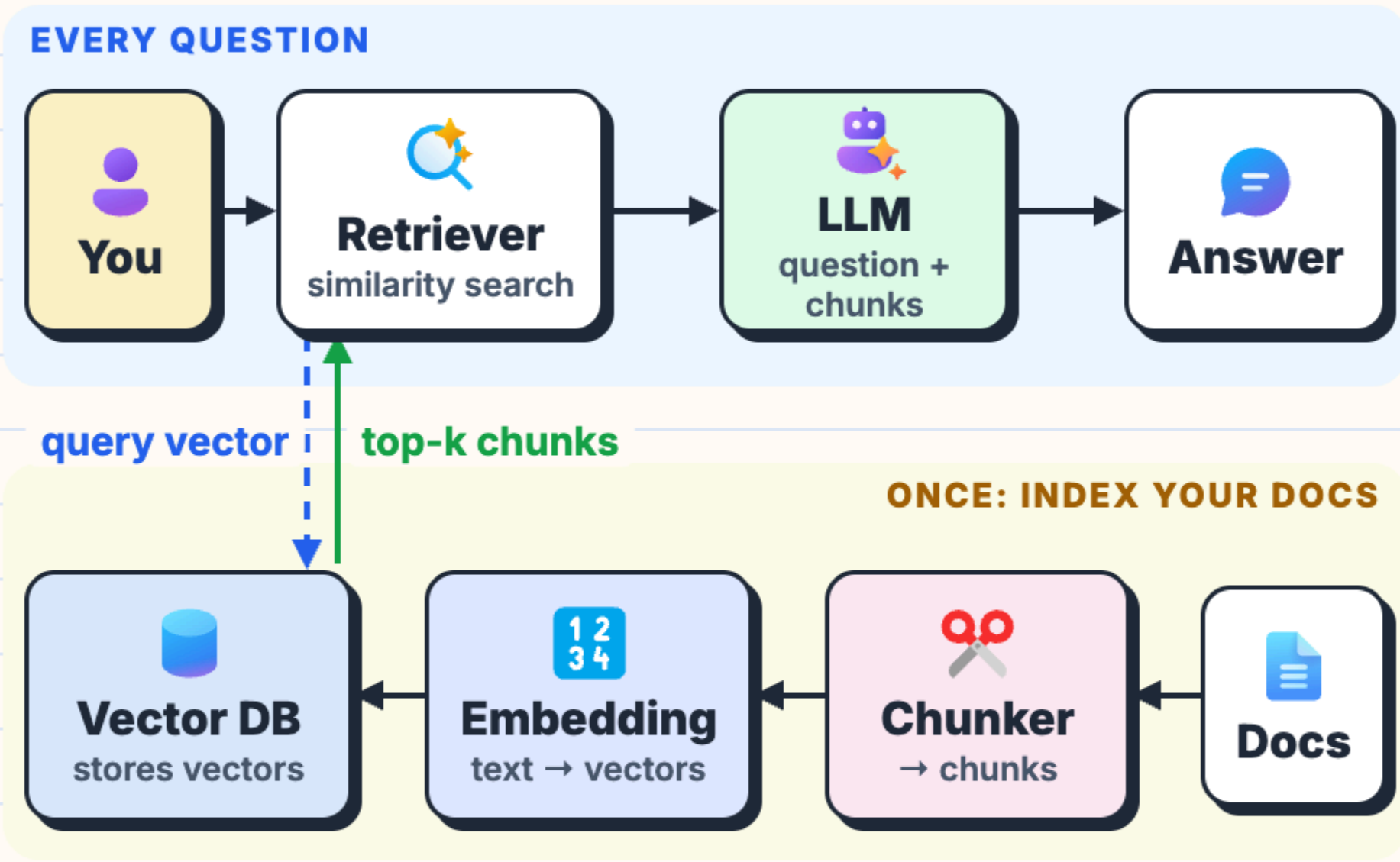
1 Naive RAG

The LLM doesn't know your data. So fetch it first, then ask.

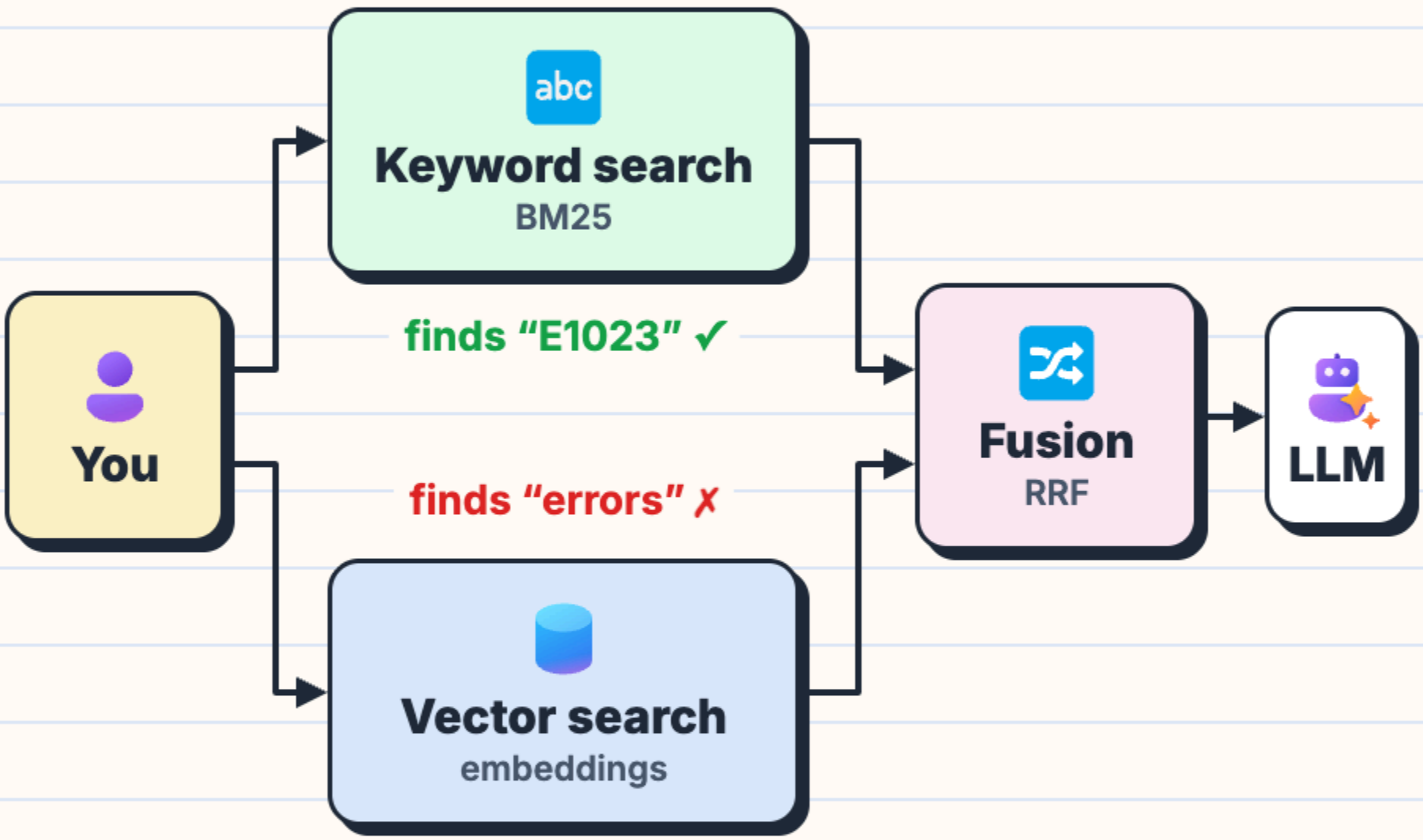


Use when: you're building v1. Chat with your company docs.

Breaks when search returns the wrong chunks. Nothing checks them. The next 7 fix that →

2 Hybrid RAG

Vector search **misses exact terms**: codes, names, IDs.

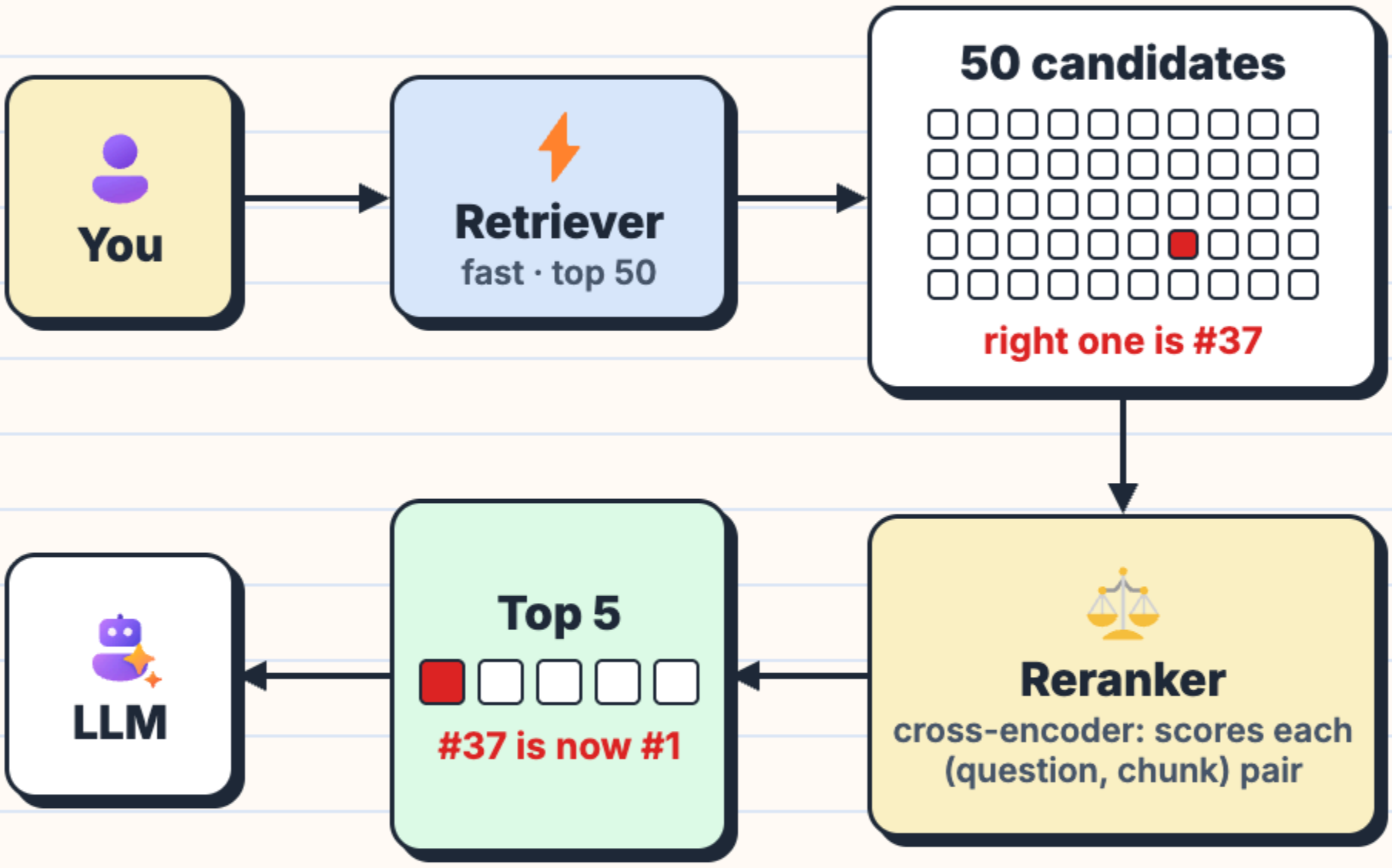


Use when: users search exact terms: codes, SKUs, names.

*Search “error E1023”: vectors find errors in general. BM25 finds **E1023**. Run both, fuse the rankings.*

3 Rerank

The right chunk was retrieved, but ranked #37. The LLM only sees the top 5.

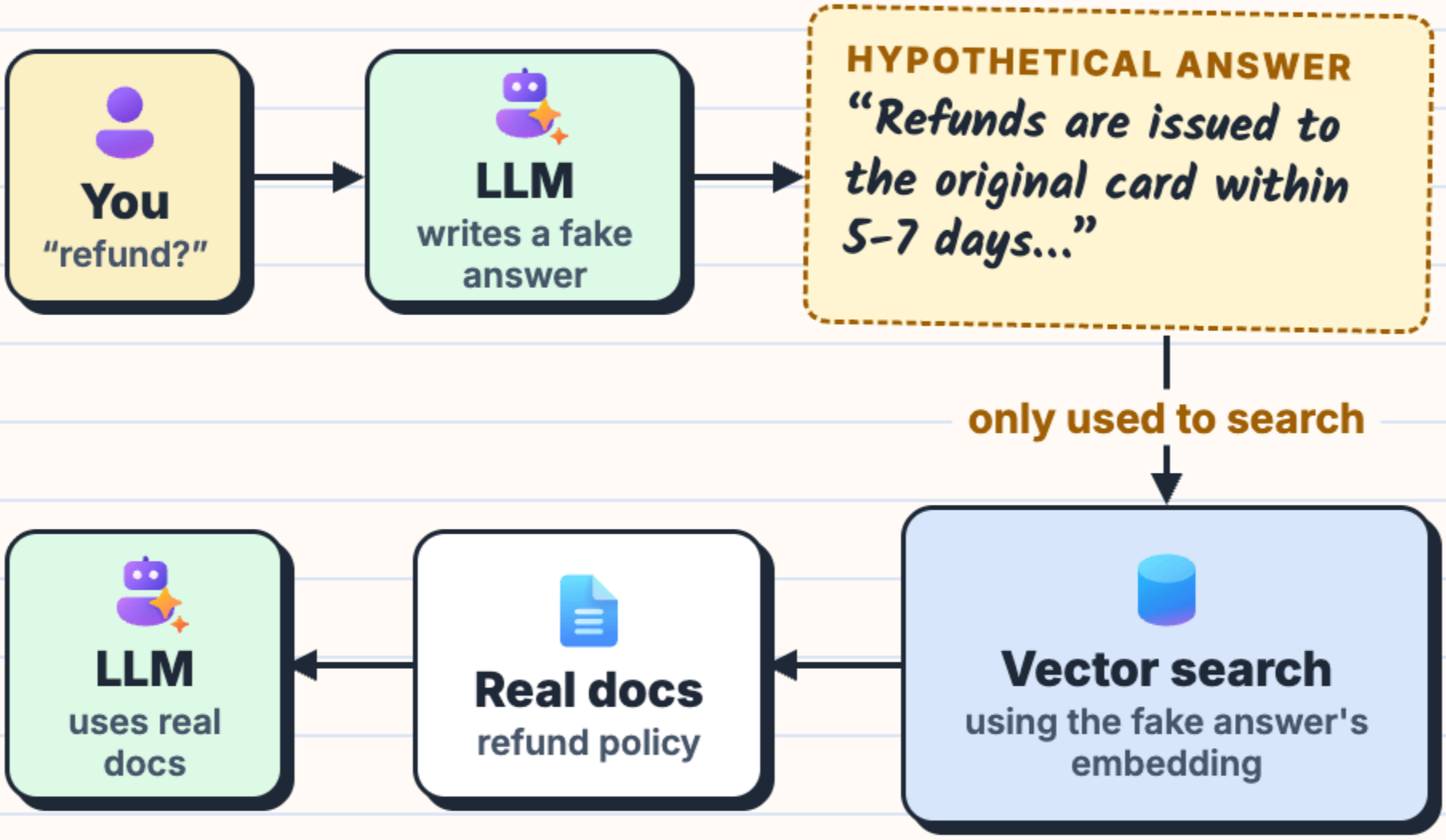


Use when: answers are **almost** right. The info is in your docs, but the LLM never sees it.

A fast retriever casts a wide net. A slower, sharper cross-encoder picks the best 5.

4 HyDE

Questions are short. Docs are long. They don't look alike, so vector search misses.

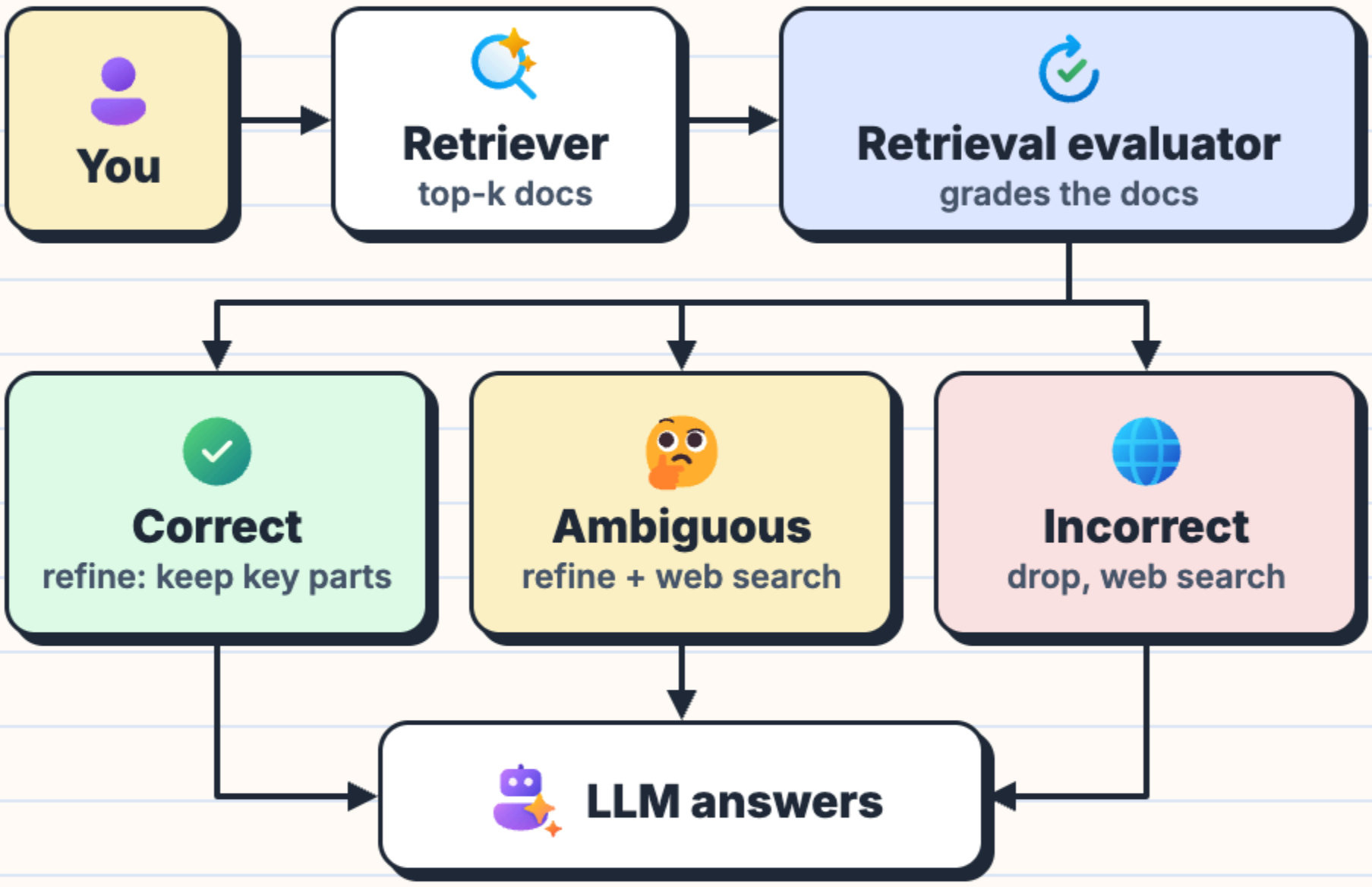


Use when: questions are short or vague, docs are long.

The fake answer can be wrong. It only has to look like the right doc. The final answer comes from real docs.

5 Corrective RAG

Retrieval can fail silently. CRAG grades the docs before the LLM sees them.

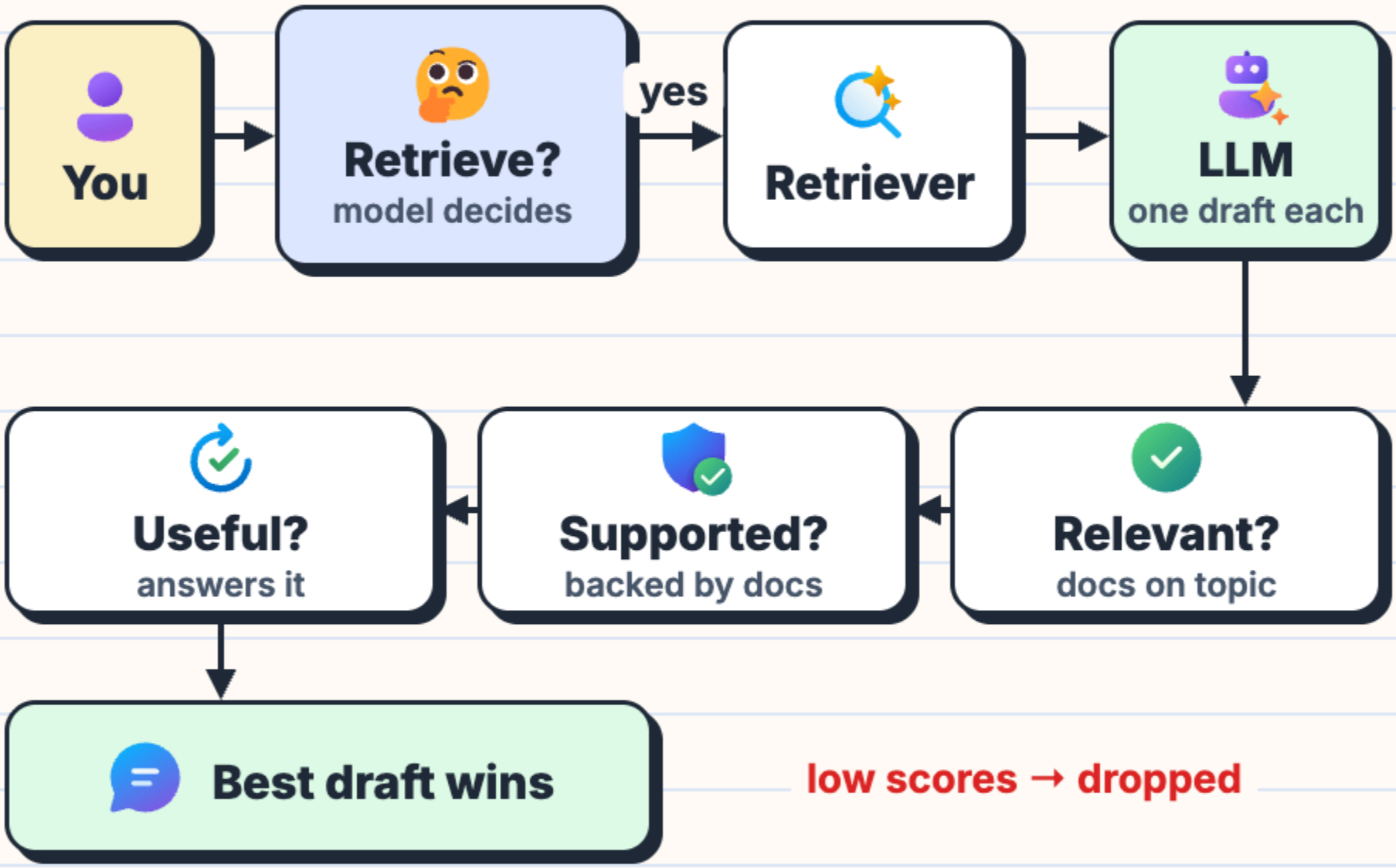


Use when: your docs have gaps and wrong answers are costly.

Docs look off-topic? Don't guess. Throw them out and search the web instead.

6 Self-RAG

The model checks its own work: need docs? relevant? backed by them?

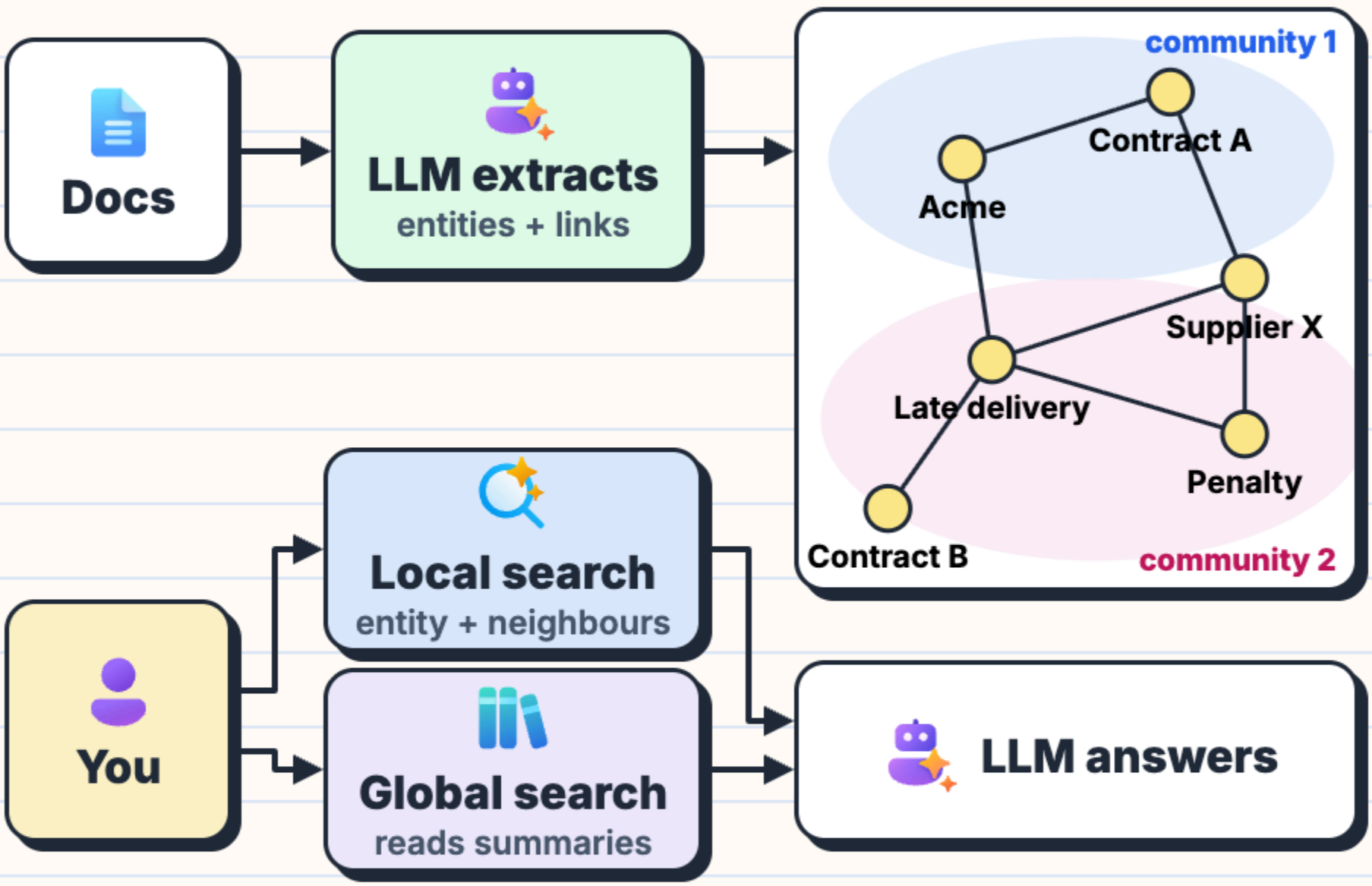


Use when: made-up answers are costly: legal, finance.

The paper trains the model to emit "reflection tokens". Most teams copy it with grader prompts.

7 GraphRAG

Some questions need connections across many docs, not 5 similar chunks.

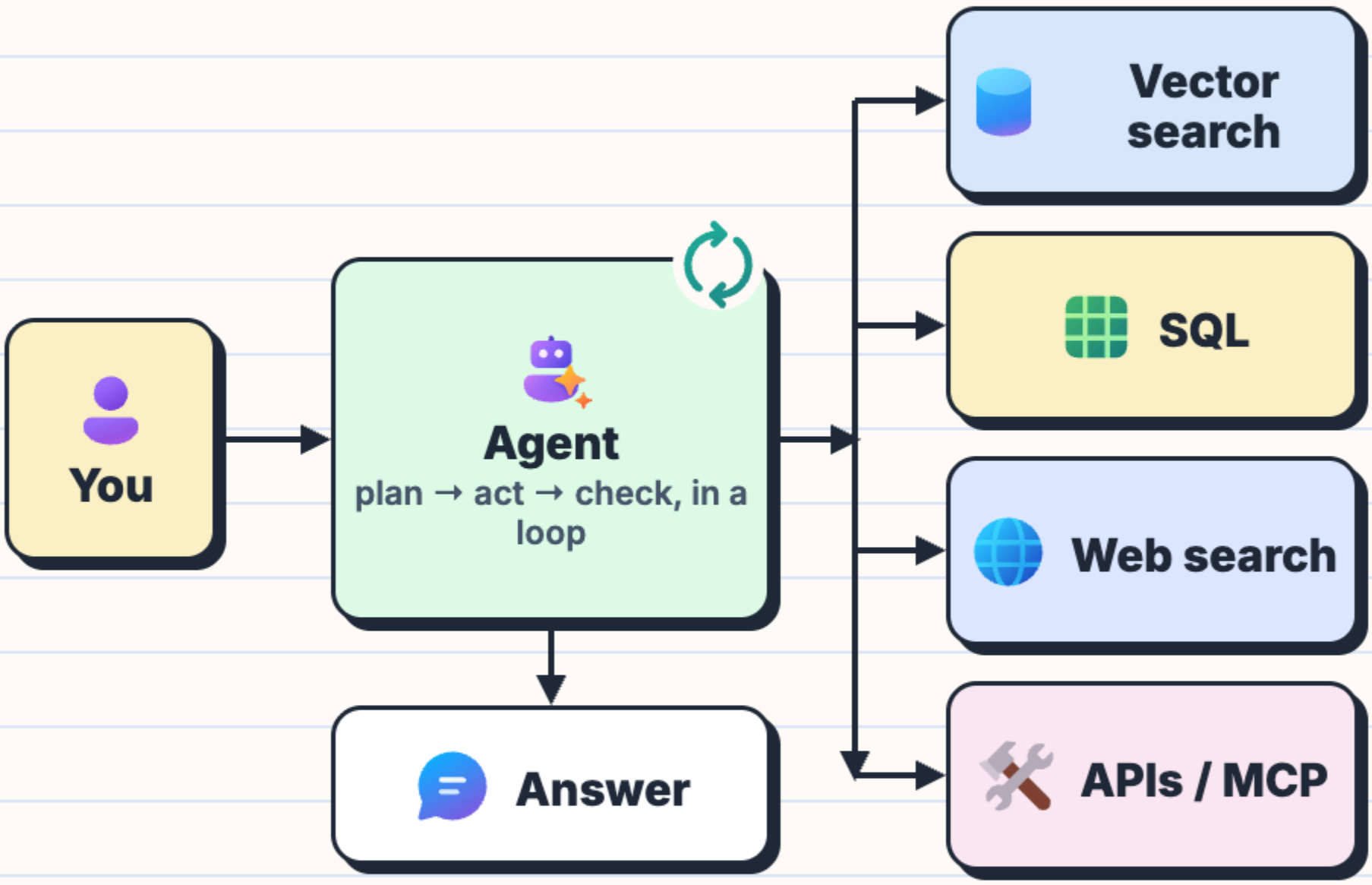


Use when: big-picture questions across many docs. Costly to index.

"Main risks across all 500 contracts?" No single chunk has that. Community summaries do.

8 Agentic RAG

Hard questions need more than one search.
An agent decides what to look up next.



Use when: multi-step questions across many sources. Slower, costlier.

"Compare Q3 sales to target and explain the gap."
SQL for numbers, docs for the target, then reason.

Which one do I pick?

Start with **Naive**. Add one fix for each problem you actually see.

Misses exact IDs, codes	**Hybrid**
Right doc found, ranked too low	**Rerank**
Short, vague questions	**HyDE**
Docs don't cover the question	**CRAG**
Confident, made-up answers	**Self-RAG**
Questions across the whole corpus	**GraphRAG**
Many steps, many sources	**Agentic**